MIL-C-37261
18 June 1976

MILITARY SPECIFICATION

CALCIUM CARBONATE TABLETS, N.F.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Calcium Carbonate Tablets, N.F.

1.2 Classification. The tablets shall be of the following sizes (see 6.2):

Size 1 - Individually Sealed, 60s
Size 2 - Bottle of 500s

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals (see 6.3).

STANDARDS

FEDERAL

FED-STD-140 - Tablets (for Medicinal Purposes).

FSC 6505

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204).

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

3. REQUIREMENTS

3.1 Material. Shall be Calcium Carbonate Tablets, N.F., and shall be in accordance with all applicable requirements of FED-STD-140 and and specified herein:

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S2. Classification. Shall be type I, class 1.

S5.2 The following is added to this paragraph:

Tablets shall contain 420 mg of precipitated calcium carbonate ($CaCO_3$) per tablet, within the applicable assay limits when determined using the N.F. method and 180 mg of aminoacetic acid (H_2NCH_2COOH) per tablet, within the designated assay limits for the tablets.

Tablets shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of aminoacetic acid, when assayed as specified in 4.3.1.

Tablets shall have an acid consuming capacity of not less than 80.0 ml of 0.1N HCl per tablet when determined as specified in 4.3.2.

Tablets shall be sugar free when determined as specified in 4.3.3.

Tablets shall contain not more than 1 mg of sodium when determined as specified in 4.3.4.

Tablets shall have little or no large gritty particles (not more than 50 mg of residue remaining on 40 mesh screen) when determined as specified in 4.3.5.

S6.4.3 Flavored tablets. The tablets shall be pleasantly flavored, free of grittiness, and shall be palatable and pleasant to the taste with no unpleasant after-taste. The tablets shall comply with the palatability test specified in 4.3.6.

S6.4.6 Hardness. The average hardness of 10 tablets shall be not less than 5.0 Kg and not more than 8.0 Kg when tested using the Pfizer Hardness Tester or another tester giving equivalent results. In addition, the hardness of each of the 10 tablets shall be not less than 4.5 Kg. If one, but not more than one, tablet has a hardness of less than 4.5 Kg, 20 additional tablets shall be tested. Not more than 2 of the 30 tablets shall have a hardness of less than 4.5 Kg.

S6.4.10.1 Uncoated tablets. Tablets shall meet the requirements of the Weight Variation Test for Tablets, as set forth in the U.S.P., including any supplements or revisions thereto.

3.2 Palatability. Not later than the time specified for opening the bids or receipt of proposals, the offeror shall submit to the contracting officer 6 units of product for size 1 and 3 units of product for size 2 which are representative of the product which the offeror proposes to furnish. A portion of the samples will be subjected to panel testing for a determination of palatability (see 4.3.6 Palatability test). A portion will be used by the

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cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval as to palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the specification requirement.

3.3 Ingredients.

3.3.1 Precipitated Calcium Carbonate and Aminoacetic Acid. The Precipitated Calcium Carbonate and Aminoacetic Acid used in the manufacture of the tablets shall comply with the tests, standards, and requirements of the U.S.P.

3.3.2 Other ingredients. All other ingredients used in the manufacture of the tablets shall comply with S5.1 of FED-STD-140.

3.4 Expiration dating period. The expiration dating period shall be not less than 36 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period (potency period) specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

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4.2 Sampling. The sampling as shown in FED-STD-140 shall be applicable.

4.3 Tests.

4.3.1 Assay for Aminoacetic Acid in Tablets.

Weigh and finely powder not less than 20 tablets. Accurately weigh approximately 450 mg of the pulverized tablets into a 30 ml, glass-stoppered centrifuge tube. Add 20.0 ml of purified water. Place on a mechanical shaker for 30 minutes. Filter through Whatman No. 3 paper and place 10.0 ml of the filtrate in a 50 ml beaker. Place 10.0 ml of purified water in a beaker as a blank. Adjust the sample solution to pH 6.0 with 0.1 N HCl. Add 5.0 ml formaldehyde (37 percent solution) and titrate to pH 9.0 with standard 0.1 N NaOH. To the purified water blank add 5.0 ml formaldehyde (37 percent solution) and titrate to pH 9.0 with standard 0.1 N NaOH.

Calculations:

% Required amount of aminoacetic acid/tablet =

$$\frac{(V_{Sample} - V_{Blank}) \times \text{Normality of NaOH} \times 75.1 \times 2 \times \text{average tablet wt} \times 100}{\text{Sample Weight} \times 180}$$

4.3.2 Acid Consuming Capacity.

Reagents.

Hydrochloric acid standard, 1.0N
Sodium Hydroxide standard, 0.1N
Methyl red indicator solution - prepare by dissolving 100 mg of methyl red in 100 ml of 95% ethanol.

Procedure.

Place 1 tablet in a 50 ml beaker. Pipet 15.00 ml of 1.0N HCl into the beaker. Pipet 1.00 ml of 1.0N HCl into a 50 ml erlenmeyer flask. Add a stirring bar to the sample and stir mechanically for 15 minutes, then pipet 1.00 ml of this sample solution to 50 ml erlenmeyer flask. Treat the sample and blank as follows: a) add 10 ml of purified water, b) add 3 drops of methyl red indicator solution, c) titrate with 0.1N sodium hydroxide to the complete disappearance of pink color and record the volume used. Calculate the acid consuming capacity for the sample.

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Calculation:

Acid consuming capacity = (150) (N) (Vb - Vs)
(ml of 0.1N HCl/tab)

Where: N = the exact normality of the standard 0.1N NaOH.
Vs = the volume of standard 0.1N NaOH used for sample.
Vb = the volume of standard 0.1N NaOH used for blank.

4.3.3 Absence of reducing and non-reducing sugars in tablets.

Finely powder not less than 20 tablets. Weigh approximately 1.5 Gram of the pulverized tablets into a glass-stoppered centrifuge tube. Add 15 ml of purified water. Place on a mechanical shaker for 15 minutes. Filter the mixture and add 1.0 ml of the filtrate to 5.0 ml of hot alkaline cupric tartrate test solution, U.S.P. (Fehling's Solution). The absence of any red-colored precipitation shall indicate the absence of reducing sugars.

Place 10.0 ml of filtrate into a 25 ml flask and add 0.5 ml of concentrated hydrochloric acid. Test the resulting solution with pHydrion paper; pH shall be less than 2.0. Heat the solution on a steam table for 15 minutes. Add 1.0 ml of this solution to 5.0 ml of hot alkaline cupric tartrate test solution, U.S.P. The absence of any red-colored precipitation shall indicate the absence of non-reducing sugars.

4.3.4 Assay for sodium contents in tablets.

Preparation of standards

Accurately weigh 2.542 Gram of sodium chloride analyzed reagent, previously dried at 140°C., and quantitatively transfer to a 1000 ml volumetric flask. Bring to volume with purified water. From this stock solution, accurately prepare two standard solutions. The concentrations of the standard solutions shall be 2.5 mg and 5.0 mg of sodium per liter.

Operating conditions

The following operating conditions shall be employed when a Beckman DU Spectrophotometer equipped with an electronic power supply and a flame photometry accessory is used. If another suitable instrument is used, the operating conditions may be modified, in order to give comparable results.

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Power Supply:

Sensitivity - No. 4 position.
Zero suppression - Off

Spectrophotometer:

Sensitivity - Approximately 1/2 turn from clockwise limit.
Slit width - 0.015 mm.
Selector switch - 0.1 position
Load resistor - No. 1
Wavelength - 589 mu 1/

Gas pressure:

Hydrogen - 3.5 p.s.i.
Oxygen - 10.0 p.s.i.

Procedure.

Weigh and finely powder not less than 20 tablets. Accurately weigh approximately 200 mg of the pulverized tablets into a 500 ml volumetric flask. Bring to volume with purified water and stir magnetically for two hours. Allow any solids which are present to settle and then, decant off the turbid liquid. Determine the wavelength at which maximum response to sodium emission is obtained (approximately 589 mu) using the standard solutions.

With the maximum wavelength selected, introduce purified water into the atomizer, set the transmission dial at zero (0) and null the meter. Next, spray the 2.5 mg/liter sodium standard into the flame, set the transmission dial at fifty (50) percent and null the meter by adjusting the sensitivity. Immediately feed the sample into the atomizer and null the meter using the transmission knob. Record the percent transmission. The 5.0 mg/liter sodium standard is then introduced and the readings noted. Repeat the above procedure with the standards and each sample a sufficient number of times to secure a reliable reading for each solution.

Construct a calibration curve (percent transmission versus sodium concentration) from the blank and the two standards. Determine the sodium concentration of the sample from the curve.

1/ The exact wavelength setting, which may be slighly different than 589 mu, shall be determined from the maximum response to sodium emission.

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Calculate the number of milligrams of sodium per tablet using the following formula:

mg sodium/tablet =

$$\frac{\text{sodium conc. of sample (mg/liter) x average tablet weight x 0.5}}{\text{Sample weight (ml)}}$$

4.3.5 Quantity of residue (gritty particles) per tablet.

Weigh, to the nearest 1/10 of a milligram, a clear, dry section of 40 mesh, stainless steel screen, approximately 3-1/2 inches in diameter. Place a single tablet into a 35-ml glass-stoppered centrifuge tube. Add 20 ml of purified water, stopper, and shake mechanically (3 inch stroke, 200 to 300 cycles per minute) for 5 minutes. Wet the stainless steel screen with purified water. Filter the solution through the tared screen and collect the filtrate in a 600 ml beaker. Rinse the centrifuge tube with three 20-ml portions of purified water and wash the residue on the screen with each of these rinsings. Dry the screen at 105°C for 15 minutes, allow to cool, and reweigh. Test four (4) tablets from each lot.

Calculate the residue remaining on the screen.

4.3.6 Palatability Test.

A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing (samples will be tested undiluted), coded, and served to panel members under controlled service conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be

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rated equal to or better than the Palatability Standard 1/, when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the samples shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

1/ The Palatability Standard is available upon separate request to the contracting officer, Defense Personnel Support Center, Directorate of Medical Materiel.

4.4 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.2.1 Unit of issue. One box containing 60 individually sealed size 1 tablets or one bottle containing 500 size 2 tablets, as specified, constitutes one unit of issue.

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5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container for size 1 tablets shall be a commercially available blister or strip package and the immediate container for size 2 tablets shall be a commercially available bottle. The immediate containers shall be considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2.2.2 Unit package. Sixty individually packaged size 1 tablets shall be packaged as specified in 5.2.5 of PPP-C-186 except that partitions are not required. At the option of the contractor, a unit package may be furnished for size 2 tablets. If a unit package is furnished, each bottle shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve unit packages containing size 1 tablets shall be packaged as specified in 5.2.7.1 of PPP-C-186 and twelve units of size 2 tablets shall be packaged as specified in 5.2.9 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.3.1 Level A. Twelve intermediate packages (144 units) of size 1 tablets or two intermediate packages (24 units) of size 2 tablets shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Twelve intermediate packages (144 units) of size 1 tablets shall be packed in an exterior container as specified in 5.4.3.1.1 of PPP-C-186. Two intermediate packages (24 units) of size 2 tablets shall be packed in an exterior container as specified in 5.4.3.2 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container.

5.4.1.1.1 Size 1. Each immediate container shall be labeled in accordance with commercial practice.

5.4.1.1.2 Size 2. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the expiration date.

5.4.1.2 Unit packages. Each unit package shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the National Stock Number.

(c) the expiration date (size 1 only).

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

(4) If a unit package is not supplied for size 2 tablets, the National Stock Number shall appear on the immediate container label.

5.4.2 Intermediate package for sizes 1 and 2. Each intermediate package shall be marked in accordance with paragraph 5.5.3 of PPP-C-186. Type II Shelf Life Markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° F (59° - 86°F)".

5.4.3 Exterior container for sizes 1 and 2. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Type II Shelf Life Markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° F (59° - 86° F)".

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6. NOTES

6.1 Ordering Data. Procurement documents should specify the following:

(a) Title Number and date of this specification.

(b) The National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock Number	Item Identification	Specification Designation
6505-00-133-5422	CALCIUM CARBONATE TABLETS, NF, 0.42 Gram, Individually Sealed, 60s	Size 1
6505-00-890-1658	CALCIUM CARBONATE TABLETS, NF, 0.42 Gram, 500s	Size 2

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1413

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE: CALCIUM CARBONATE TABLETS, NF MIL-C-37261

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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